

Section 6A. Model Selection and Regularization

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Recap: Linear Models

- ▶ Consider a linear model of the form

$$y = \beta_0 + \sum_{j=1}^p \beta_j X_j + \varepsilon = f_L(X; \beta) + \varepsilon$$

- ▶ We find the coefficients β from data $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ by solving:

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^N (y_i - f_L(\mathbf{x}_i; \beta))^2$$

- ▶ In many problems, we have a **very large number of inputs** ($p \gg 1$); hence,
 - ▶ The model can be *hard to interpret*
 - ▶ The prediction accuracy can be low due to *high variance*

Feature Selection Problem: Select a subset of the most relevant inputs

Feature Selection

We have three main methods for feature/input selection:

- ▶ ***Subset selection***: Identify the subset of inputs most relevant to predict the output
- ▶ ***Regularization methods***: Shrink some coefficients β_j towards zero, effectively removing the effect of the corresponding inputs
- ▶ ***Dimensionality reduction***: Generate $M < p$ artificial inputs using the information in the original p inputs and use these artificial inputs predict the output

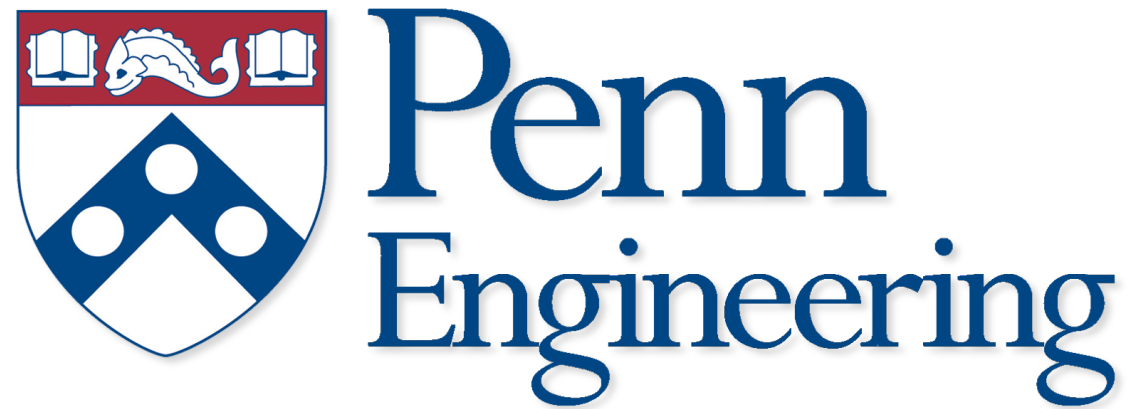
Best Subset Selection

Problem: Given p inputs, find a subset of inputs most relevant to explain the output

Algorithm:

1. $\mathcal{M}_0$ denotes the null model $\hat{y} = \hat{\beta}_0$ with no inputs
2. For $k = 1$ to p , do the following:
 - 2.1 Fit all $\binom{p}{k}$ models with exactly k inputs (chosen from p possible inputs)
 - 2.2 Out of all those $\binom{p}{k}$ models, pick the one with the smallest (training) RSS, $\mathcal{M}_k$
3. End For-loop
4. Out of the $p + 1$ models $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$, find the model that minimizes the cross-validated **testing error**

Issue: This algorithm requires the analysis of 2^p models, which is prohibiting when p is large



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